

React Router Configuration

1. App.js

jsx

```
import React from 'react';
import { BrowserRouter, Routes, Route, Link, NavLink, Navigate } from 'react-router-dom';
import './App.css';

// Simple Components
import Home from './pages/Home';
import About from './pages/About';
import Users from './pages/Users';
import UserProfile from './pages/UserProfile';
import Dashboard from './pages/Dashboard';
import Login from './pages/Login';
import NotFound from './pages/NotFound';

function App() {
  return (
    <BrowserRouter>
      <div className="app">
        { /* Navigation Bar */ }
        <nav className="navbar">
          <h2 className="logo">🌐 Router Demo</h2>
          <div className="nav-links">
            <NavLink to="/" className={({isActive}) => isActive ? 'active' : ''>Home</NavLink>
            <NavLink to="/about" className={({isActive}) => isActive ? 'active' : ''>About</NavLink>
            <NavLink to="/users" className={({isActive}) => isActive ? 'active' : ''>Users</NavLink>
            <NavLink to="/dashboard" className={({isActive}) => isActive ? 'active' : ''>Dashboard</NavLink>
          </div>
        </nav>

        { /* Routes Configuration */ }
        <div className="content">
          <Routes>
            <Route path="/" element={ <Home /> } />
            <Route path="/about" element={ <About /> } />
            <Route path="/users" element={ <Users /> } />
            <Route path="/users/:id" element={ <UserProfile /> } />
            <Route path="/dashboard" element={ <Dashboard /> } />
            <Route path="/login" element={ <Login /> } />
            <Route path="/404" element={ <NotFound /> } />
            <Route path="*" element={ <Navigate to="/404" replace /> } />
          </Routes>
        </div>

        { /* Router Info Footer */ }
        <div className="router-info">
          <p>React Router v6 | Current Path: {window.location.pathname}</p>
        </div>
      </div>
    </BrowserRouter>
  );
}

export default App;
```

2. App.css

CSS

```
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

body {
  font-family: -apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto, sans-serif;
  background: #f0f2f5;
}

.app {
```

```
min-height: 100vh;
display: flex;
flex-direction: column;
}

/* Navbar */
.navbar {
  background: white;
  padding: 1rem 2rem;
  display: flex;
  justify-content: space-between;
  align-items: center;
  box-shadow: 0 2px 8px rgba(0,0,0,0.1);
}

.logo {
  color: #4263eb;
  font-size: 1.5rem;
}

.nav-links {
  display: flex;
  gap: 1.5rem;
}

.nav-links a {
  text-decoration: none;
  color: #495057;
  font-weight: 500;
  padding: 0.5rem 1rem;
  border-radius: 4px;
  transition: all 0.3s;
}

.nav-links a:hover {
  background: #e7f5ff;
  color: #4263eb;
}

.nav-links a.active {
  background: #4263eb;
  color: white;
}

/* Content Area */
.content {
  flex: 1;
  padding: 2rem;
  max-width: 1200px;
  margin: 0 auto;
  width: 100%;
}

/* Cards */
.card {
  background: white;
  border-radius: 8px;
  padding: 1.5rem;
  box-shadow: 0 2px 8px rgba(0,0,0,0.1);
  margin-bottom: 1rem;
}

.card h1 {
  color: #212529;
  margin-bottom: 1rem;
}

.card p {
  color: #495057;
  line-height: 1.6;
  margin-bottom: 1rem;
}

/* Buttons */
.btn {
  background: #4263eb;
  color: white;
  border: none;
```

```
padding: 0.5rem 1rem;
border-radius: 4px;
cursor: pointer;
font-size: 1rem;
margin-right: 0.5rem;
text-decoration: none;
display: inline-block;
}

.btn:hover {
  background: #364fc7;
}

.btn.secondary {
  background: #868e96;
}

/* User List */
.user-list {
  list-style: none;
  margin-top: 1rem;
}

.user-list li {
  padding: 0.75rem;
  border: 1px solid #dee2e6;
  border-radius: 4px;
  margin-bottom: 0.5rem;
}

.user-list a {
  text-decoration: none;
  color: #4263eb;
  font-weight: 500;
}

/* Router Info Footer */
.router-info {
  background: #212529;
  color: #adb5bd;
  padding: 1rem;
  text-align: center;
  font-size: 0.9rem;
}
```

3. pages/Home.js

jsx

```
import React from 'react';
import { Link } from 'react-router-dom';

function Home() {
  return (
    <div className="card">
      <h1> Home Page</h1>
      <p>Welcome to React Router Configuration Demo!</p>

      <div style={{ marginTop: '2rem' }}>
        <h3>Router Configuration:</h3>
        <pre style={{ background: '#f8f9fa', padding: '1rem', borderRadius: '4px' }}>
          {`
<BrowserRouter>
  <Routes>
    <Route path="/" element={<Home />} />
    <Route path="/about" element={<About />} />
    <Route path="/users" element={<Users />} />
    <Route path="/users/:id" element={<UserProfile />} />
    <Route path="/dashboard" element={<Dashboard />} />
    <Route path="/login" element={<Login />} />
    <Route path="*" element={<Navigate to="/404" />} />
  </Routes>
</BrowserRouter>
`}
        </pre>
      </div>

      <Link to="/about" className="btn">Learn More →</Link>
    </div>
  );
}
```

```

    </div>
  );
}

export default Home;

```

4. pages/About.js

```

jsx

import React from 'react';
import { useNavigate } from 'react-router-dom';

function About() {
  const navigate = useNavigate();

  return (
    <div className="card">
      <h1>    About Page</h1>
      <p>This is a static route: <code>/about</code></p>

      <h3>Navigation Examples:</h3>
      <button className="btn" onClick={() => navigate('/')}>Go Home</button>
      <button className="btn secondary" onClick={() => navigate(-1)}>Go Back</button>

      <div style={{ marginTop: '1rem' }}>
        <p><strong>useNavigate Hook:</strong> Programmatic navigation</p>
        <pre style={{ background: '#f8f9fa', padding: '0.5rem' }}>
          const navigate = useNavigate();
          navigate('/'); // Go to home
          navigate(-1); // Go back
        </pre>
      </div>
    </div>
  );
}

export default About;

```

5. pages/Users.js

```

jsx

import React from 'react';
import { Link, useSearchParams } from 'react-router-dom';

function Users() {
  const [searchParams, setSearchParams] = useSearchParams();
  const users = [
    { id: 1, name: 'John Doe' },
    { id: 2, name: 'Jane Smith' },
    { id: 3, name: 'Bob Johnson' }
  ];

  const filter = searchParams.get('filter') || '';

  return (
    <div className="card">
      <h1>    Users Page</h1>

      <div style={{ marginBottom: '1rem' }}>
        <input
          type="text"
          placeholder="Filter users..."
          value={filter}
          onChange={(e) => setSearchParams({ filter: e.target.value })}
          style={{ padding: '0.5rem', width: '200px' }}
        />
      </div>

      <h3>Dynamic Routes with useParams:</h3>
      <ul className="user-list">
        {users
          .filter(user => user.name.toLowerCase().includes(filter.toLowerCase()))
          .map(user => (
            <li key={user.id}>
              <Link to={`/users/${user.id}`}>{user.name}</Link>
            </li>
          ))}
      </ul>
    </div>
  );
}

```

```

        </li>
      )}
    </ul>

    <div style={{ marginTop: '1rem' }}>
      <p><strong>Query Params:</strong> {searchParams.toString() || 'None'}</p>
      <pre style={{ background: '#f8f9fa', padding: '0.5rem' }}>
        const [searchParams] = useSearchParams();
        const filter = searchParams.get('filter');
      </pre>
    </div>
  </div>
);
}

export default Users;

```

6. pages/UserProfile.js

```

jsx

import React from 'react';
import { useParams, Link } from 'react-router-dom';

function UserProfile() {
  const { id } = useParams();
  const users = [
    1: { name: 'John Doe', email: 'john@example.com', role: 'Admin' },
    2: { name: 'Jane Smith', email: 'jane@example.com', role: 'User' },
    3: { name: 'Bob Johnson', email: 'bob@example.com', role: 'Editor' }
  ];

  const user = users[id];

  return (
    <div className="card">
      <h1> User Profile</h1>

      {user ? (
        <>
          <p><strong>Name:</strong> {user.name}</p>
          <p><strong>Email:</strong> {user.email}</p>
          <p><strong>Role:</strong> {user.role}</p>
          <p><strong>URL Parameter:</strong> <code>:id = {id}</code></p>
        </>
      ) : (
        <p>User not found</p>
      )}

      <pre style={{ background: '#f8f9fa', padding: '0.5rem', marginTop: '1rem' }}>
        const {'{ id }'} = useParams();
        // URL: /users/{id}
      </pre>

      <Link to="/users" className="btn secondary">← Back to Users</Link>
    </div>
  );
}

export default UserProfile;

```

7. pages/Dashboard.js

```

jsx

import React, { useState } from 'react';
import { Navigate } from 'react-router-dom';

function Dashboard() {
  const [isAuthenticated, setIsAuthenticated] = useState(false);

  if (!isAuthenticated) {
    return <Navigate to="/login" replace />;
  }

  return (
    <div className="card">

```

```

    <h1>    Dashboard (Protected Route)</h1>
    <p>This route is protected. You can only see this if authenticated.</p>

    <button className="btn danger" onClick={() => setIsAuthenticated(false)}>
      Logout
    </button>

    <div style={{ marginTop: '1rem' }}>
      <h3>Protected Route Pattern:</h3>
      <pre style={{ background: '#f8f9fa', padding: '0.5rem' }}>
        {`
function ProtectedRoute({ children }) {
  if (!isAuthenticated) {
    return <Navigate to="/login" replace />;
  }
  return children;
}

// Usage in Routes:
<Route
  path="/dashboard"
  element={
    <ProtectedRoute>
      <Dashboard />
    </ProtectedRoute>
  }
/>
        `
      </pre>
    </div>
  </div>
);
}

export default Dashboard;

```

8. pages/Login.js

```

jsx

import React from 'react';
import { useNavigate, Link } from 'react-router-dom';

function Login() {
  const navigate = useNavigate();

  const handleLogin = () => {
    // Simulate login
    navigate('/dashboard');
  };

  return (
    <div className="card" style={{ maxWidth: '400px', margin: '2rem auto' }}>
      <h1>    Login Page</h1>

      <div style={{ marginTop: '1rem' }}>
        <input
          type="text"
          placeholder="Username"
          style={{ width: '100%', padding: '0.5rem', marginBottom: '1rem' }}
        />
        <input
          type="password"
          placeholder="Password"
          style={{ width: '100%', padding: '0.5rem', marginBottom: '1rem' }}
        />
        <button className="btn" onClick={handleLogin} style={{ width: '100%' }}>
          Login
        </button>
      </div>

      <p style={{ marginTop: '1rem' }}>
        <Link to="/"><- Back to Home</Link>
      </p>
    </div>
  );
}

```

```
export default Login;
```

9. pages/NotFound.js

```
jsx

import React from 'react';
import { Link } from 'react-router-dom';

function NotFound() {
  return (
    <div className="card" style={{ textAlign: 'center' }}>
      <h1>404 - Page Not Found</h1>
      <p>The page you're looking for doesn't exist.</p>
      <Link to="/" className="btn">Go Home</Link>
    </div>
  );
}

export default NotFound;
```

10. index.js

```
jsx

import React from 'react';
import ReactDOM from 'react-dom/client';
import App from './App';

const root = ReactDOM.createRoot(document.getElementById('root'));
root.render(
  <React.StrictMode>
    <App />
  </React.StrictMode>
);
```

Output Screenshots (Text Format)

Home Page Output:

```
text

-----
| 🏠 Router Demo | Home | About | Users | Dashboard |
-----

HOME PAGE
=====
Welcome to React Router Configuration Demo!

ROUTER CONFIGURATION:
-----
<BrowserRouter>
  <Routes>
    <Route path="/" element={<Home />} />
    <Route path="/about" element={<About />} />
    <Route path="/users" element={<Users />} />
    <Route path="/users/:id" element={<UserProfile />} />
    <Route path="/dashboard" element={<Dashboard />} />
    <Route path="/login" element={<Login />} />
    <Route path="*" element={<Navigate to="/404" />} />
  </Routes>
</BrowserRouter>

[Learn More →]

-----
React Router v6 | Current Path: /
```

About Page Output:

text

ABOUT PAGE

=====

This is a static route: /about

NAVIGATION EXAMPLES:

[Go Home] [Go Back]

useNavigate Hook:

```
const navigate = useNavigate();
navigate('/'); // Go to home
navigate(-1); // Go back
```

React Router v6 | Current Path: /about

Users Page Output:

text

USERS PAGE

=====

[Filter users..._____]

DYNAMIC ROUTES WITH useParams:

- John Doe (/users/1)
- Jane Smith (/users/2)
- Bob Johnson (/users/3)

Query Params: None

```
const [searchParams] = useSearchParams();
const filter = searchParams.get('filter');
```

React Router v6 | Current Path: /users

User Profile Page Output:

text

USER PROFILE

=====

Name: John Doe
Email: john@example.com
Role: Admin
URL Parameter: :id = 1

```
const { id } = useParams();
// URL: /users/1
```

[← Back to Users]

React Router v6 | Current Path: /users/1

Dashboard Page (Protected) Output:

text

DASHBOARD (PROTECTED ROUTE)
=====

This route is protected. You can only see this if authenticated.

[Logout]

PROTECTED ROUTE PATTERN:

```
function ProtectedRoute({ children }) {  
  if (!isAuthenticated) {  
    return <Navigate to="/login" replace />;  
  }  
  return children;  
}
```

React Router v6 | Current Path: /dashboard

Login Page Output:

text

LOGIN PAGE
=====

[Username

[Password

[Login]

← Back to Home

React Router v6 | Current Path: /login

404 Page Output:

text

404 - PAGE NOT FOUND
=====

The page you're looking for doesn't exist.

[Go Home]

React Router v6 | Current Path: /non-existent

Key Router Features Demonstrated:

1. **Basic Routes** - Home, About (static pages)
2. **Dynamic Routes** - /users/:id with useParams
3. **Query Parameters** - Filter users with useSearchParams
4. **Programmatic Navigation** - useNavigate hook
5. **Protected Routes** - Dashboard with Navigate redirect
6. **Catch-all Route** - 404 page for invalid paths

7. **Active Links** - NavLink with active state styling

8. **Nested Navigation** - Users list to profile navigation

How to Run:

1. Create React app: `npx create-react-app router-demo`
2. Install React Router: `npm install react-router-dom`
3. Replace files with the code above
4. Run: `npm start`
5. Open <http://localhost:3000>

This concise demo shows all essential React Router v6 concepts in a clean, understandable format.